

Differentiable functions – 9

Integral inequalities through derivatives

Homework

1 сентября 2026 года

Theoretical minimum. Integration by parts as the main device; Wirtinger's, Poincaré's, Hardy's and Opial's inequalities; expansion in the eigenfunctions of $-y'' = \lambda y$ with the given boundary conditions; the Cauchy–Schwarz inequality in integral form; sharp constants and extremal functions; the discrete counterparts from the first sheet on limits.

1. Prove that for $f \in C^1[0, 1]$ with $\int_0^1 f = 0$ one has $\int_0^1 f^2 \leq \frac{1}{\pi^2} \int_0^1 (f')^2$, and identify the extremal functions.
2. Prove that $\|f\|_\infty^2 \leq \int_0^1 f^2 + 2 \left(\int_0^1 f^2 \right)^{1/2} \left(\int_0^1 (f')^2 \right)^{1/2}$ for $f \in C^1[0, 1]$, and deduce the embedding $H^1 \subset C$ in dimension one.
3. Prove the refined Steklov inequality: if $f \in C^1[0, 1]$ satisfies $f(0) = f(1) = 0$ and $\int_0^1 f = 0$, then $\int_0^1 f^2 \leq \frac{1}{4\pi^2} \int_0^1 (f')^2$, and explain why the constant is the second Dirichlet eigenvalue rather than the first.
4. Let $f \in C^1[0, 1]$ with $f(0) = 0$. Prove that $\int_0^1 f^2 \leq \frac{4}{\pi^2} \int_0^1 (f')^2$ and determine the extremal function.
5. Prove the Hardy inequality in the discrete form $\sum_{n \geq 1} \left(\frac{a_1 + \dots + a_n}{n} \right)^2 \leq 4 \sum_{n \geq 1} a_n^2$ for non-negative a_n , and show that the constant 4 is sharp.
6. Let $f \in C^1[a, b]$ with $f(a) = f(b) = 0$. Prove that $\max |f| \leq \frac{1}{2} \int_a^b |f'|$ and that the constant $\frac{1}{2}$ is sharp.
7. Prove that for $f \in C^2[0, 1]$ with $f(0) = f(1) = 0$, $\int_0^1 f^2 \leq \frac{1}{\pi^4} \int_0^1 (f'')^2$.
8. Let $f \in C^1[0, \infty)$ with f and f' in L^2 . Prove that $f(x) \rightarrow 0$ as $x \rightarrow \infty$ and that $\|f\|_\infty^2 \leq 2\|f\|_2\|f'\|_2$.

9. Prove the inequality $\int_0^1 |f| \leq \frac{1}{2} \max |f'|$ for $f \in C^1[0, 1]$ with $\int_0^1 f' = 0$ and $f(0) = 0$, and find the extremal function.
10. Prove that for a convex f on $[0, 1]$ $\int_0^1 f(x) dx \leq \frac{f(0)+f(1)}{2}$ and deduce the trapezoid error estimate $|\int_0^1 f - \frac{f(0)+f(1)}{2}| \leq \frac{\|f''\|_\infty}{12}$.